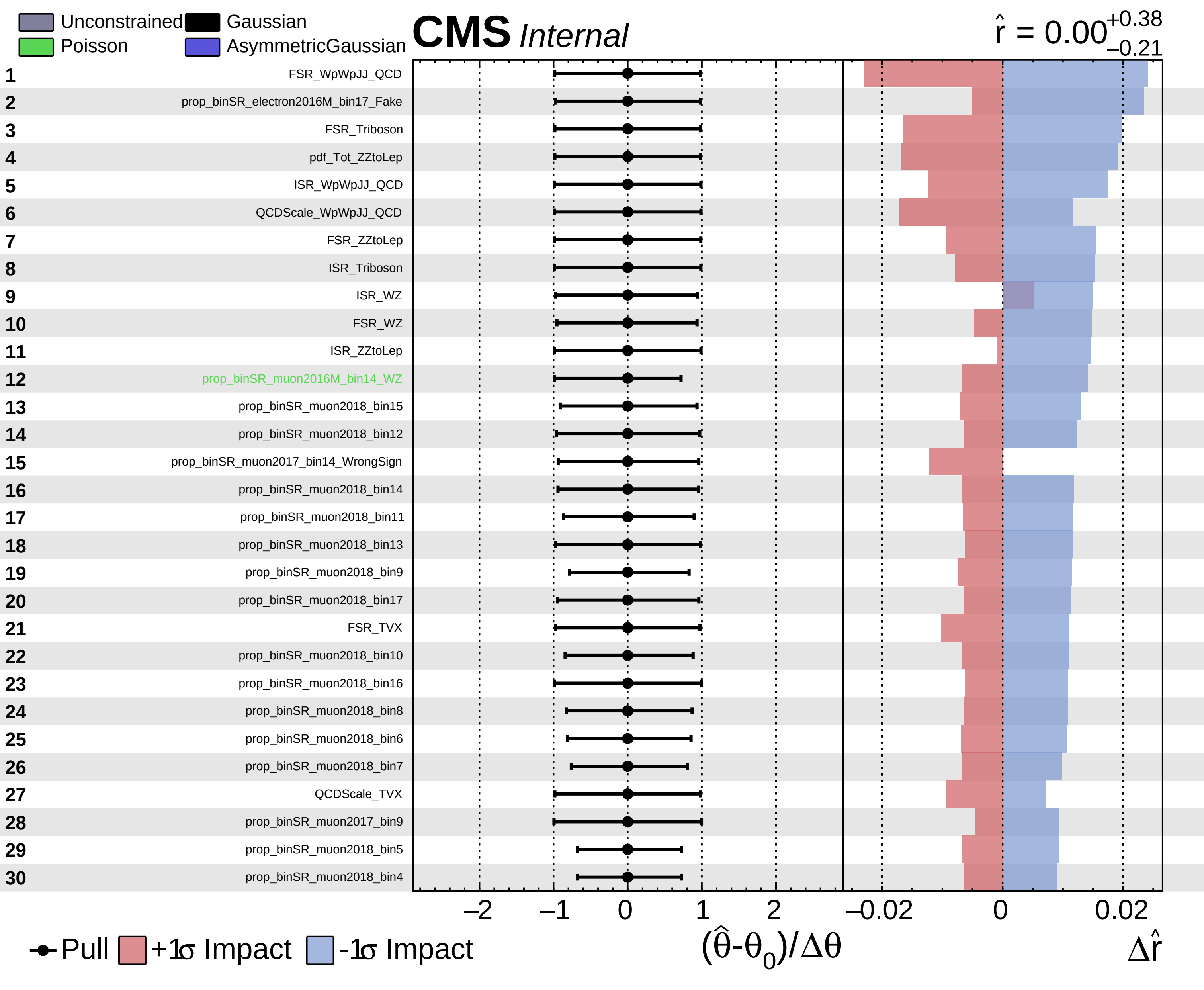
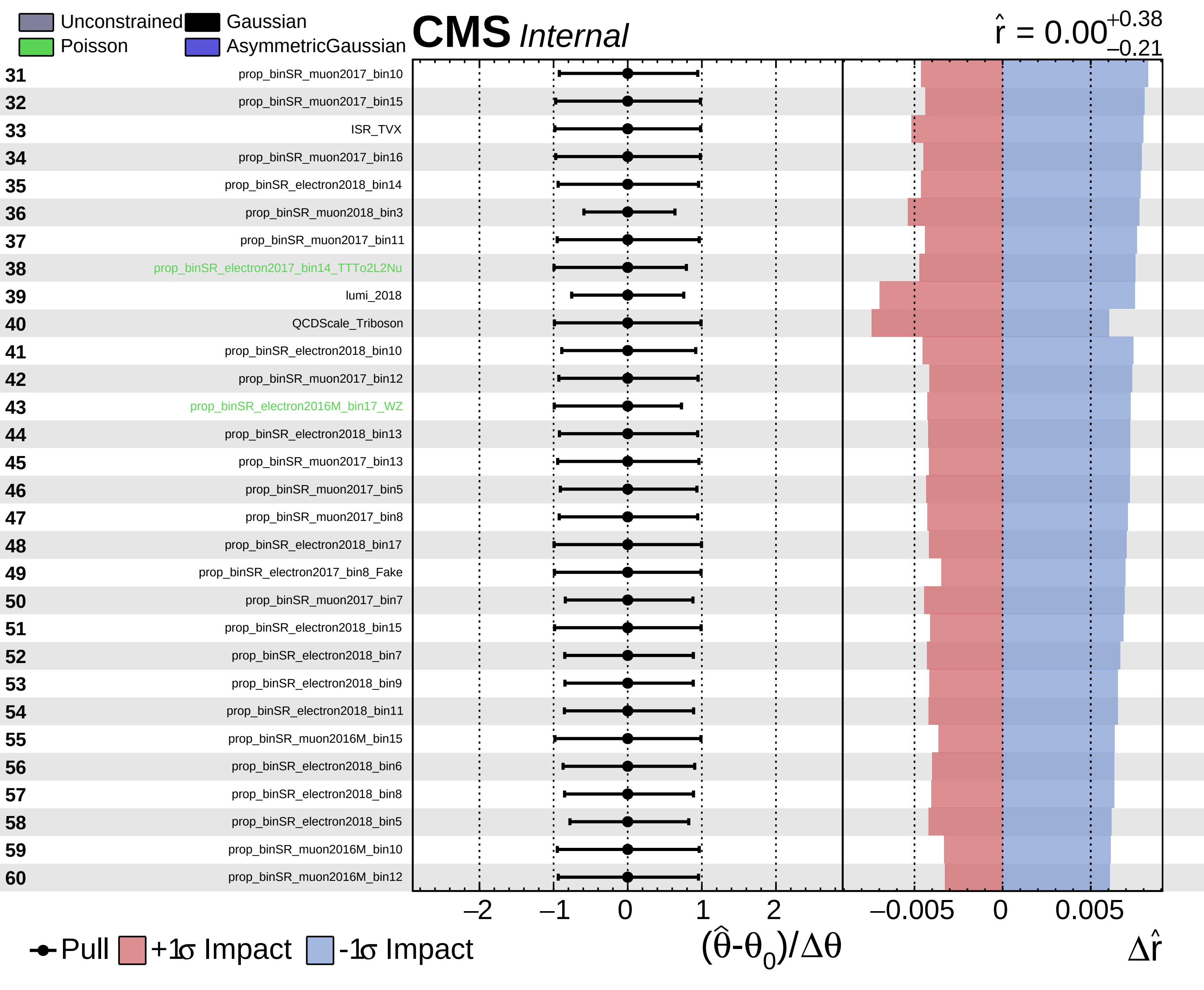
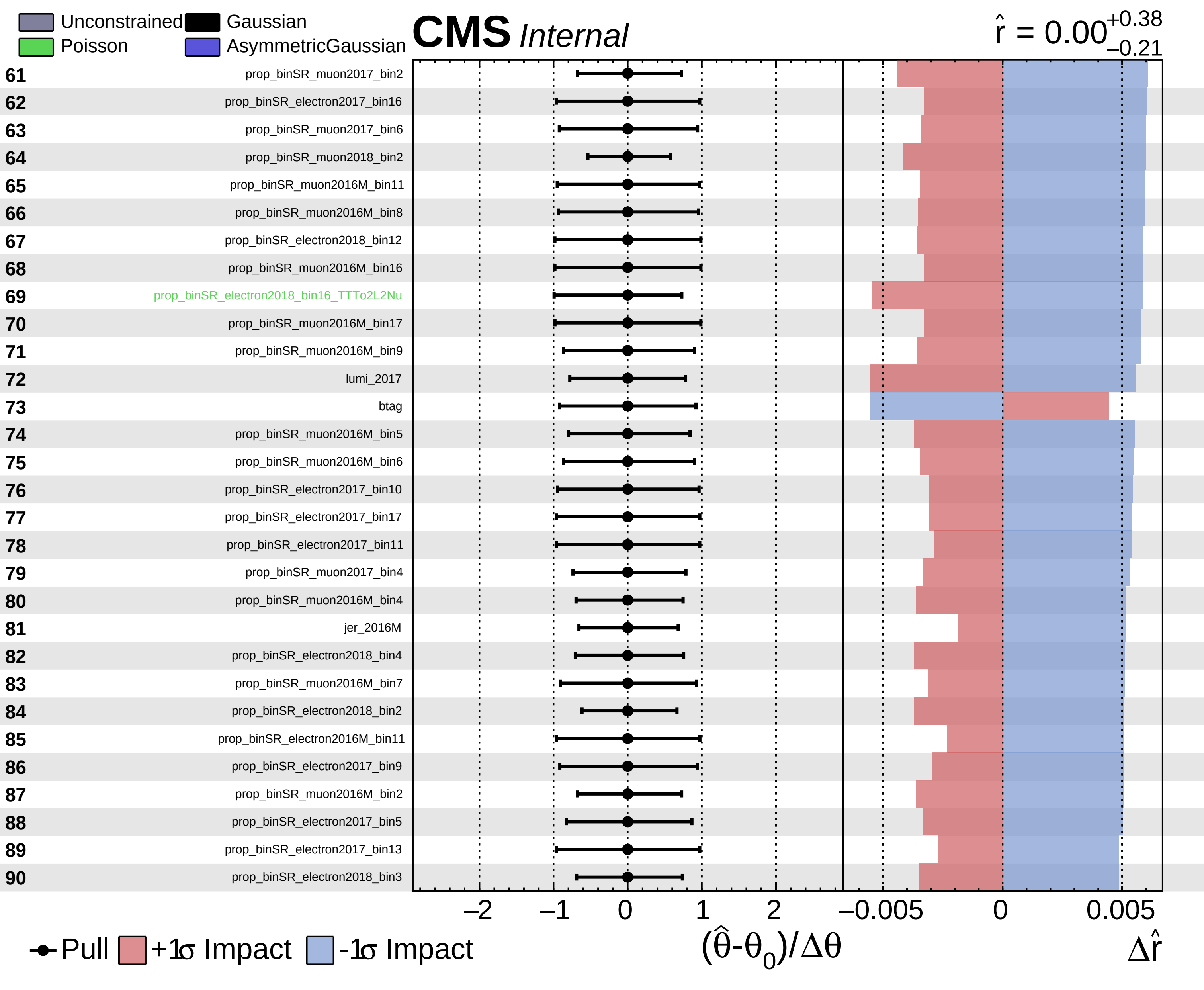
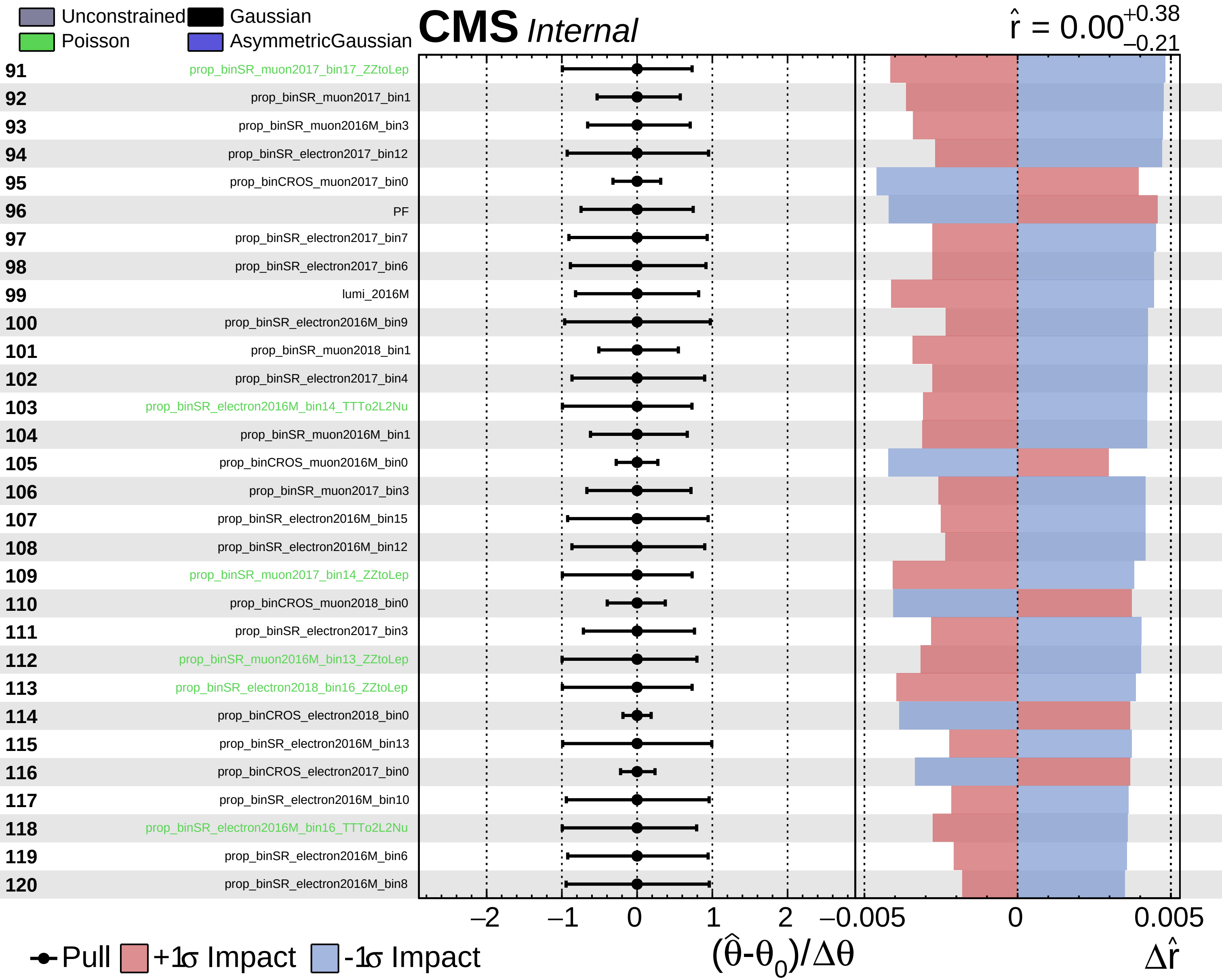


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.21}$
1 FSR_WpWpJJ_QCD
2 prop_binSR_electron2016M_bin17_Fake
3 FSR_Triboson
4 pdf_Tot_ZZtoLep
5 ISR_WpWpJJ_QCD
6 QCDScale_WpWpJJ_QCD
7 FSR_ZZtoLep
8 ISR_Triboson
9 ISR_WZ
10 FSR_WZ
11 ISR_ZZtoLep
12 prop_binSR_muon2016M_bin14_WZ
13 prop_binSR_muon2018_bin15
14 prop_binSR_muon2018_bin12
15 prop_binSR_muon2017_bin14_WrongSign
16 prop_binSR_muon2018_bin14
17 prop_binSR_muon2018_bin11
18 prop_binSR_muon2018_bin13
19 prop_binSR_muon2018_bin9
20 prop_binSR_muon2018_bin17
21 FSR_TVX
22 prop_binSR_muon2018_bin10
23 prop_binSR_muon2018_bin16
24 prop_binSR_muon2018_bin8
25 prop_binSR_muon2018_bin6
26 prop_binSR_muon2018_bin7
27 QCDScale_TVX
28 prop_binSR_muon2017_bin9
29 prop_binSR_muon2018_bin5
30 prop_binSR_muon2018_bin4
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

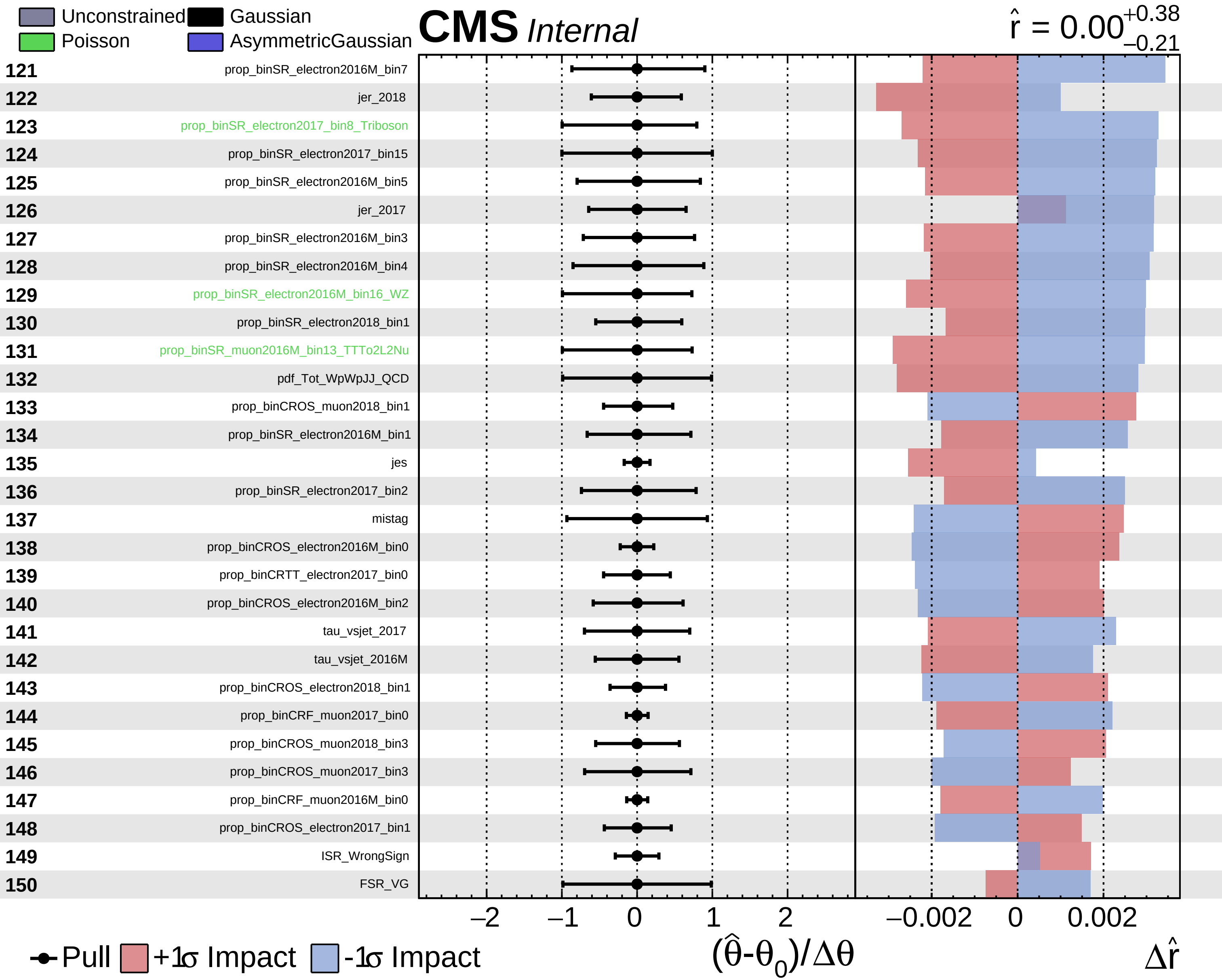

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.38 −0.21
31 prop_binSR_muon2017_bin10
32 prop_binSR_muon2017_bin15
33 ISR_TVX
34 prop_binSR_muon2017_bin16
35 prop_binSR_electron2018_bin14
36 prop_binSR_muon2018_bin3
37 prop_binSR_muon2017_bin11
38 prop_binSR_electron2017_bin14_TTTo2L2Nu
39 lumi_2018
40 QCDScale_Triboson
41 prop_binSR_electron2018_bin10
42 prop_binSR_muon2017_bin12
43 prop_binSR_electron2016M_bin17_WZ
44 prop_binSR_electron2018_bin13
45 prop_binSR_muon2017_bin13
46 prop_binSR_muon2017_bin5
47 prop_binSR_muon2017_bin8
48 prop_binSR_electron2018_bin17
49 prop_binSR_electron2017_bin8_Fake
50 prop_binSR_muon2017_bin7
51 prop_binSR_electron2018_bin15
52 prop_binSR_electron2018_bin7
53 prop_binSR_electron2018_bin9
54 prop_binSR_electron2018_bin11
55 prop_binSR_muon2016M_bin15
56 prop_binSR_electron2018_bin6
57 prop_binSR_electron2018_bin8
58 prop_binSR_electron2018_bin5
59 prop_binSR_muon2016M_bin10
60 prop_binSR_muon2016M_bin12
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂

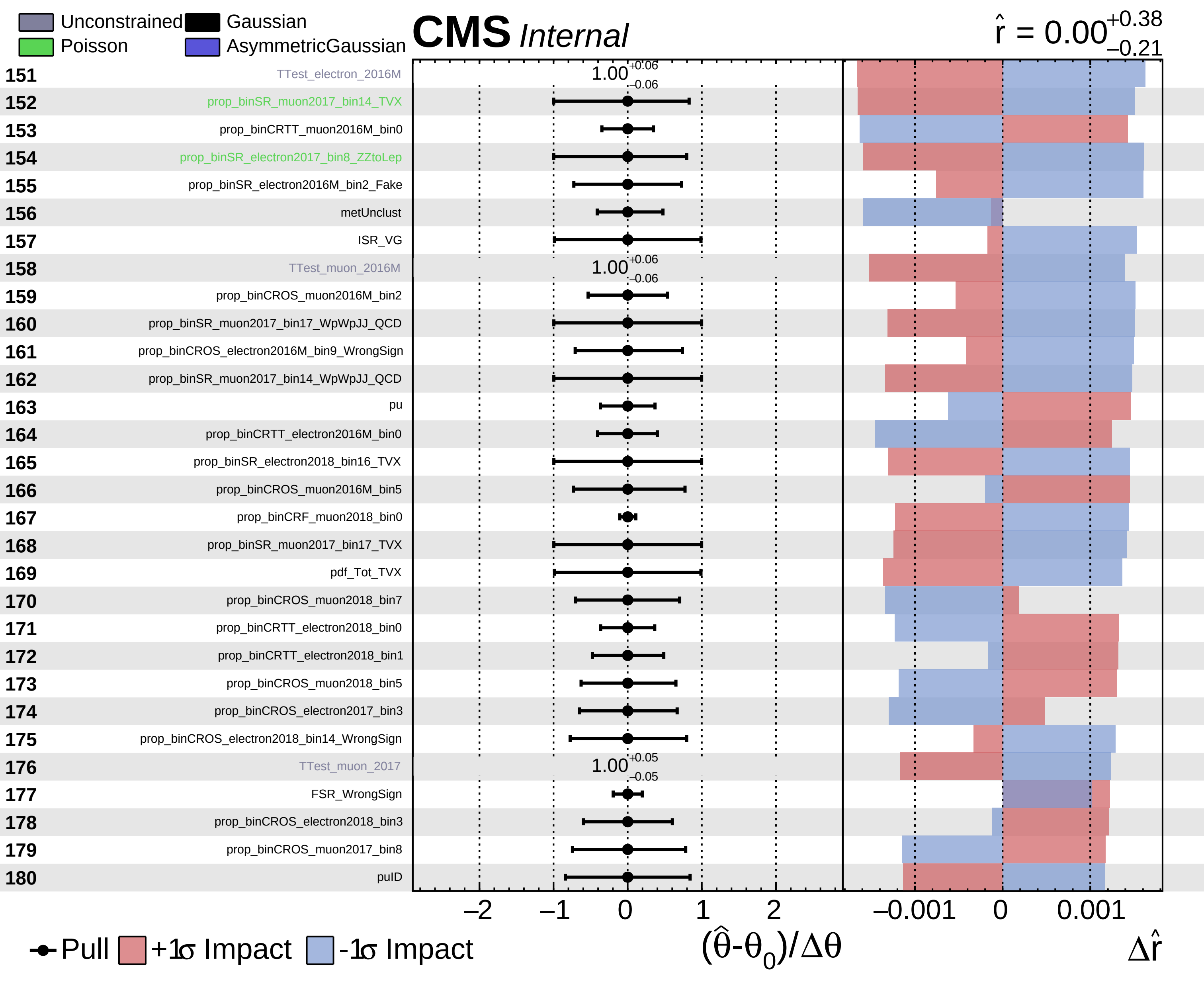


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.21}$
61 prop_binSR_muon2017_bin2
62 prop_binSR_electron2017_bin16
63 prop_binSR_muon2017_bin6
64 prop_binSR_muon2018_bin2
65 prop_binSR_muon2016M_bin11
66 prop_binSR_muon2016M_bin8
67 prop_binSR_electron2018_bin12
68 prop_binSR_muon2016M_bin16
69 prop_binSR_electron2018_bin16_TTTo2L2Nu
70 prop_binSR_muon2016M_bin17
71 prop_binSR_muon2016M_bin9
72 lumi_2017
73 btag
74 prop_binSR_muon2016M_bin5
75 prop_binSR_muon2016M_bin6
76 prop_binSR_electron2017_bin10
77 prop_binSR_electron2017_bin17
78 prop_binSR_electron2017_bin11
79 prop_binSR_muon2017_bin4
80 prop_binSR_muon2016M_bin4
81 jer_2016M
82 prop_binSR_electron2018_bin4
83 prop_binSR_muon2016M_bin7
84 prop_binSR_electron2018_bin2
85 prop_binSR_electron2016M_bin11
86 prop_binSR_electron2017_bin9
87 prop_binSR_muon2016M_bin2
88 prop_binSR_electron2017_bin5
89 prop_binSR_electron2017_bin13
90 prop_binSR_electron2018_bin3
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

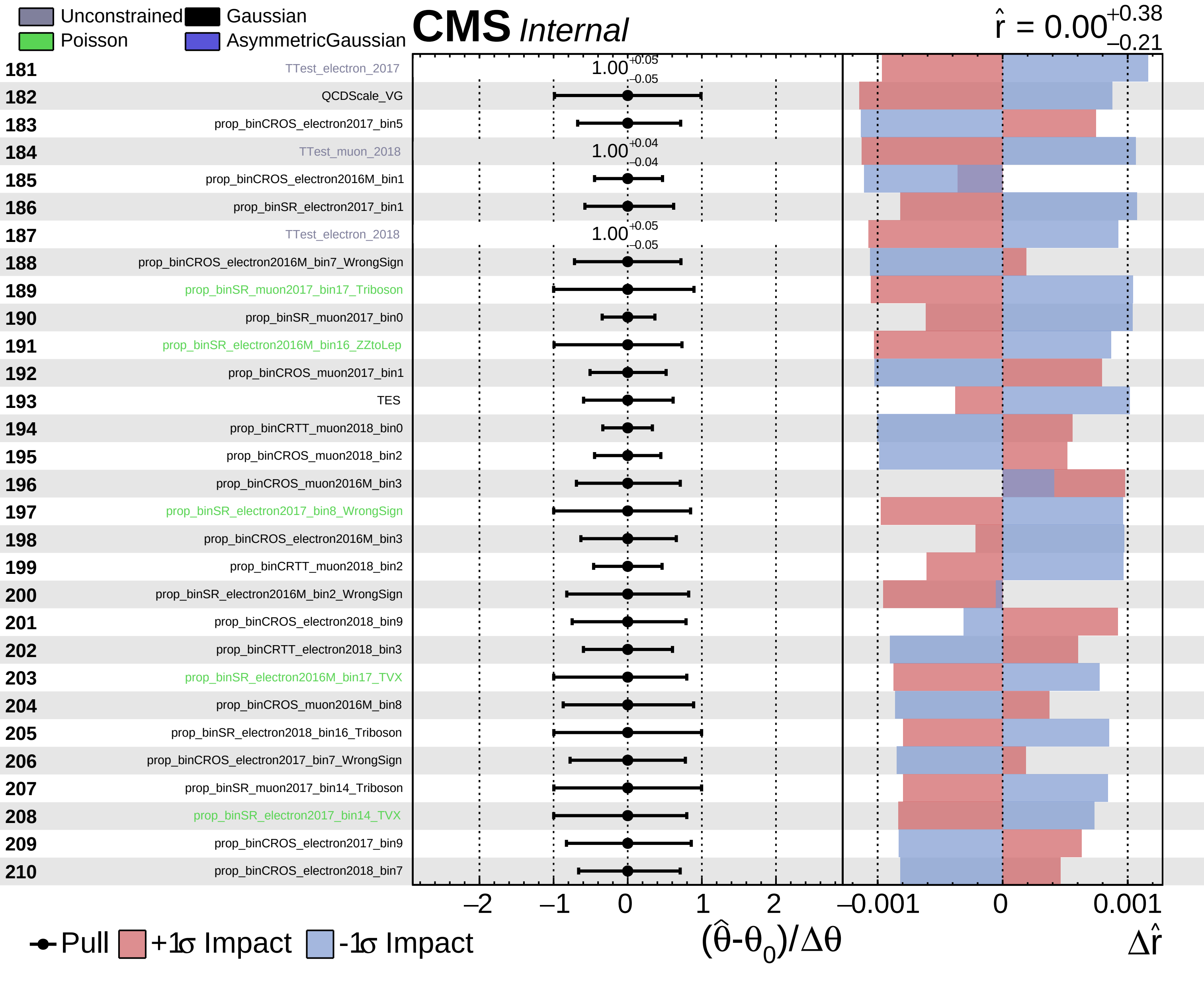


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.21}$
91 prop_binSR_muon2017_bin17_ZZtoLep
92 prop_binSR_muon2017_bin1
93 prop_binSR_muon2016M_bin3
94 prop_binSR_electron2017_bin12
95 prop_binCROS_muon2017_bin0
96 PF
97 prop_binSR_electron2017_bin7
98 prop_binSR_electron2017_bin6
99 lumi_2016M
100 prop_binSR_electron2016M_bin9
101 prop_binSR_muon2018_bin1
102 prop_binSR_electron2017_bin4
103 prop_binSR_electron2016M_bin14_TTTo2L2Nu
104 prop_binSR_muon2016M_bin1
105 prop_binCROS_muon2016M_bin0
106 prop_binSR_muon2017_bin3
107 prop_binSR_electron2016M_bin15
108 prop_binSR_electron2016M_bin12
109 prop_binSR_muon2017_bin14_ZZtoLep
110 prop_binCROS_muon2018_bin0
111 prop_binSR_electron2017_bin3
112 prop_binSR_muon2016M_bin13_ZZtoLep
113 prop_binSR_electron2018_bin16_ZZtoLep
114 prop_binCROS_electron2018_bin0
115 prop_binSR_electron2016M_bin13
116 prop_binCROS_electron2017_bin0
117 prop_binSR_electron2016M_bin10
118 prop_binSR_electron2016M_bin16_TTTo2L2Nu
119 prop_binSR_electron2016M_bin6
120 prop_binSR_electron2016M_bin8
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

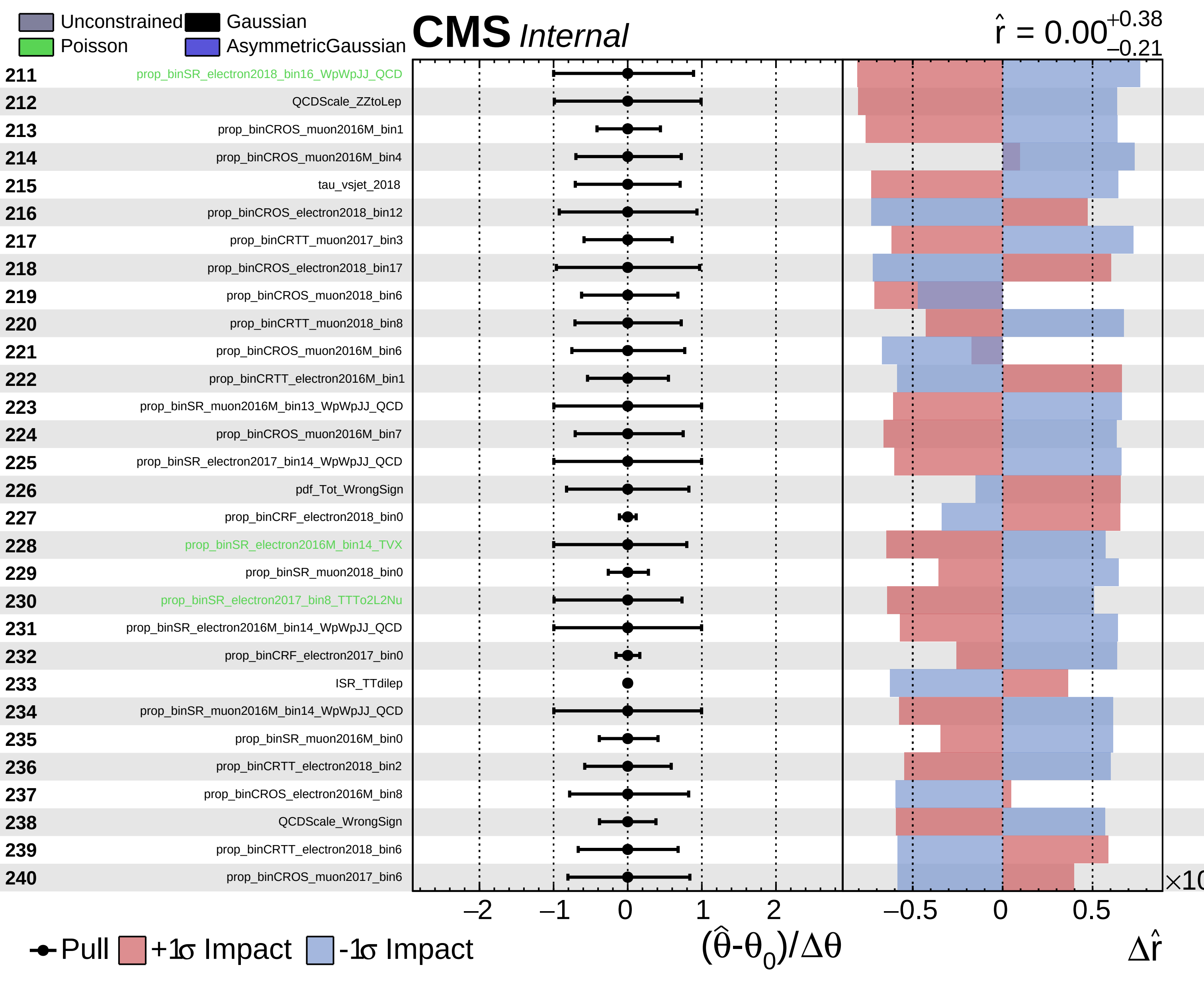

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.21}$
121 prop_binSR_electron2016M_bin7
122 jer_2018
123 prop_binSR_electron2017_bin8_Triboson
124 prop_binSR_electron2017_bin15
125 prop_binSR_electron2016M_bin5
126 jer_2017
127 prop_binSR_electron2016M_bin3
128 prop_binSR_electron2016M_bin4
129 prop_binSR_electron2016M_bin16_WZ
130 prop_binSR_electron2018_bin1
131 prop_binSR_muon2016M_bin13_TTTo2L2Nu
132 pdf_Tot_WpWpJJ_QCD
133 prop_binCROS_muon2018_bin1
134 prop_binSR_electron2016M_bin1
135 jes
136 prop_binSR_electron2017_bin2
137 mistag
138 prop_binCROS_electron2016M_bin0
139 prop_binCRTT_electron2017_bin0
140 prop_binCROS_electron2016M_bin2
141 tau_vsjet_2017
142 tau_vsjet_2016M
143 prop_binCROS_electron2018_bin1
144 prop_binCRF_muon2017_bin0
145 prop_binCROS_muon2018_bin3
146 prop_binCROS_muon2017_bin3
147 prop_binCRF_muon2016M_bin0
148 prop_binCROS_electron2017_bin1
149 ISR_WrongSign
150 FSR_VG
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

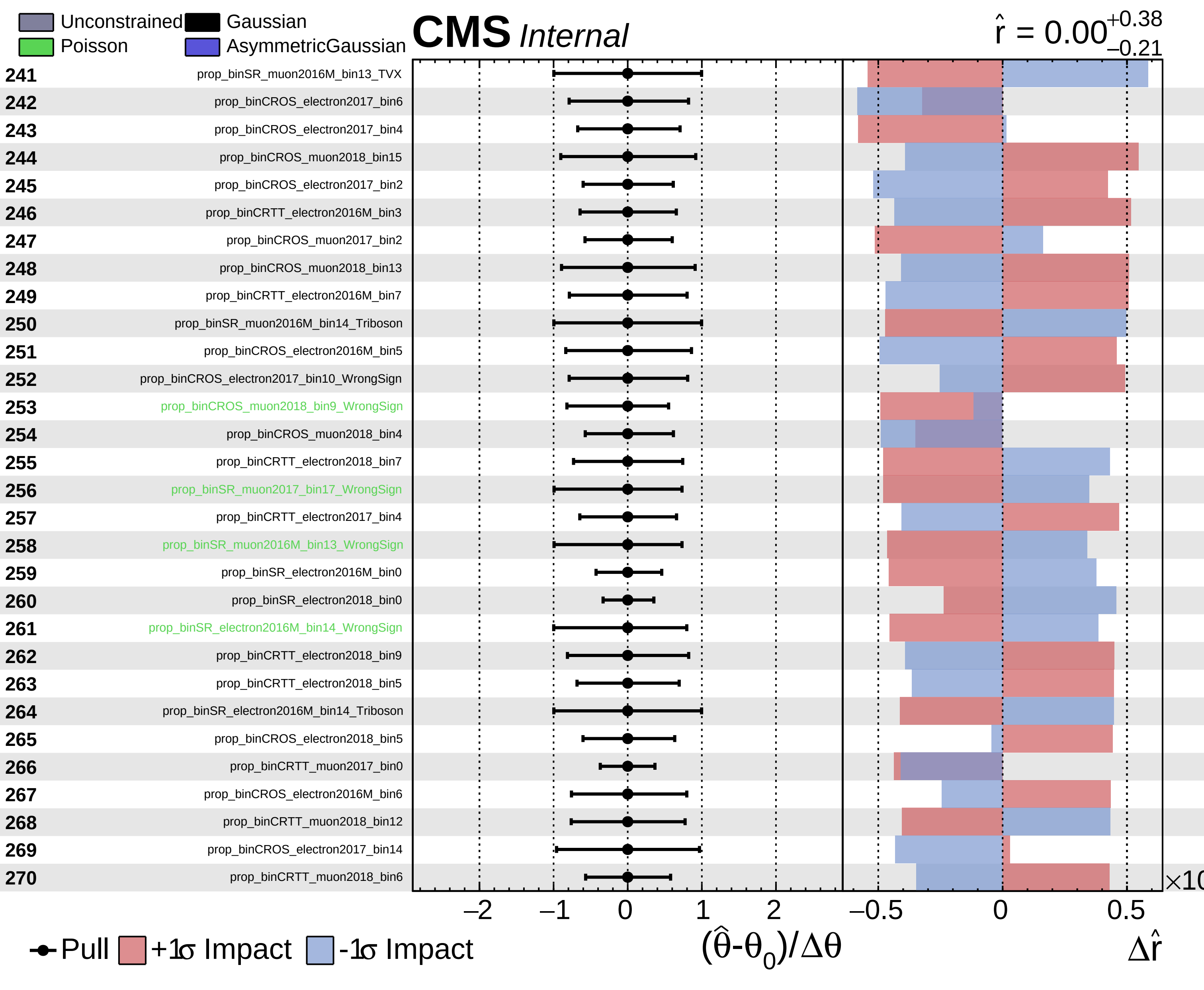


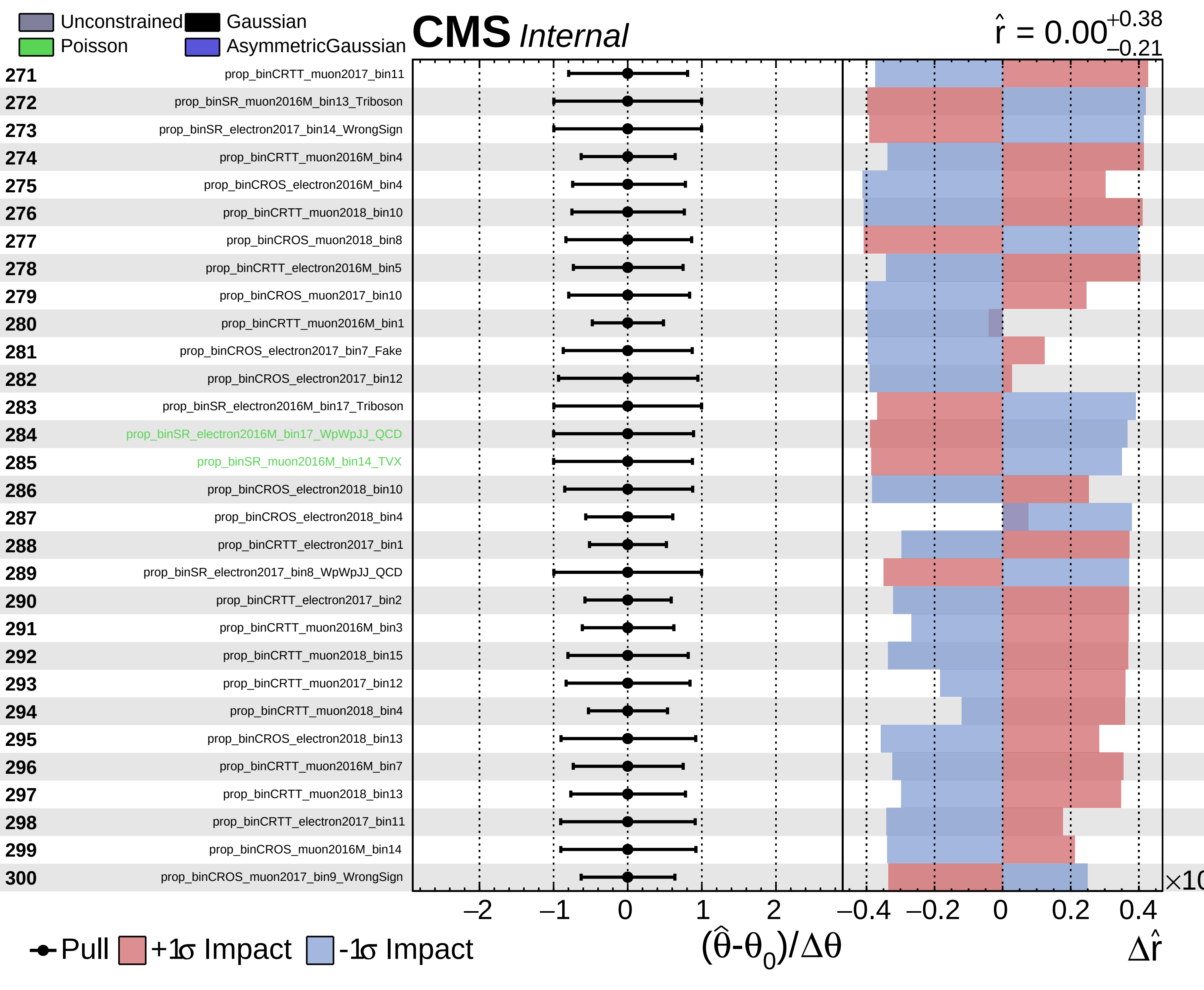
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.21}$
151 TTest_electron_2016M
$1.00^{+0.06}_{-0.06}$
152 prop_binSR_muon2017_bin14_TVX
153 prop_binCRTT_muon2016M_bin0
154 prop_binSR_electron2017_bin8_ZZtoLep
155 prop_binSR_electron2016M_bin2_Fake
156 metUnclust
157 ISR_VG
158 TTest_muon_2016M
$1.00^{+0.06}_{-0.06}$
159 prop_binCROS_muon2016M_bin2
160 prop_binSR_muon2017_bin17_WpWpJJ_QCD
161 prop_binCROS_electron2016M_bin9_WrongSign
162 prop_binSR_muon2017_bin14_WpWpJJ_QCD
163 pu
164 prop_binCRTT_electron2016M_bin0
165 prop_binSR_electron2018_bin16_TVX
166 prop_binCROS_muon2016M_bin5
167 prop_binCRF_muon2018_bin0
168 prop_binSR_muon2017_bin17_TVX
169 pdf_Tot_TVX
170 prop_binCROS_muon2018_bin7
171 prop_binCRTT_electron2018_bin0
172 prop_binCRTT_electron2018_bin1
173 prop_binCROS_muon2018_bin5
174 prop_binCROS_electron2017_bin3
175 prop_binCROS_electron2018_bin14_WrongSign
176 TTest_muon_2017
$1.00^{+0.05}_{-0.05}$
177 FSR_WrongSign
178 prop_binCROS_electron2018_bin3
179 prop_binCROS_muon2017_bin8
180 puID
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



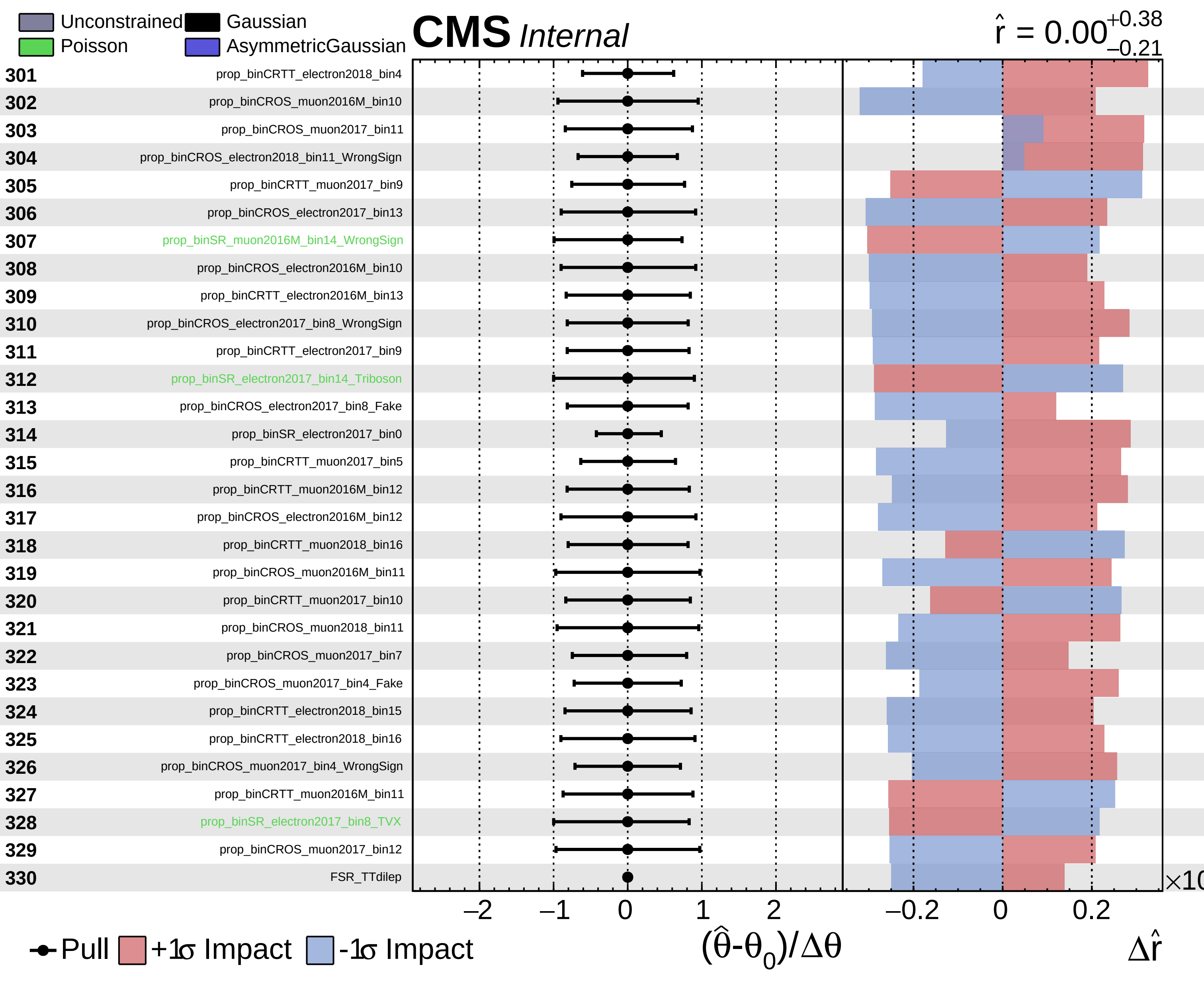
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.21}$
181 TTest_electron_2017 $1.00^{+0.05}_{-0.05}$
182 QCDScale_VG
183 prop_binCROS_electron2017_bin5
184 TTest_muon_2018 $1.00^{+0.04}_{-0.04}$
185 prop_binCROS_electron2016M_bin1
186 prop_binSR_electron2017_bin1
187 TTest_electron_2018 $1.00^{+0.05}_{-0.05}$
188 prop_binCROS_electron2016M_bin7_WrongSign
189 prop_binSR_muon2017_bin17_Triboson
190 prop_binSR_muon2017_bin0
191 prop_binSR_electron2016M_bin16_ZZtoLep
192 prop_binCROS_muon2017_bin1
193 TES
194 prop_binCRTT_muon2018_bin0
195 prop_binCROS_muon2018_bin2
196 prop_binCROS_muon2016M_bin3
197 prop_binSR_electron2017_bin8_WrongSign
198 prop_binCROS_electron2016M_bin3
199 prop_binCRTT_muon2018_bin2
200 prop_binSR_electron2016M_bin2_WrongSign
201 prop_binCROS_electron2018_bin9
202 prop_binCRTT_electron2018_bin3
203 prop_binSR_electron2016M_bin17_TVX
204 prop_binCROS_muon2016M_bin8
205 prop_binSR_electron2018_bin16_Triboson
206 prop_binCROS_electron2017_bin7_WrongSign
207 prop_binSR_muon2017_bin14_Triboson
208 prop_binSR_electron2017_bin14_TVX
209 prop_binCROS_electron2017_bin9
210 prop_binCROS_electron2018_bin7
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.21}$
211 prop_binSR_electron2018_bin16_WpWpJJ_QCD
212 QCDScale_ZZtoLep
213 prop_binCROS_muon2016M_bin1
214 prop_binCROS_muon2016M_bin4
215 tau_vsjet_2018
216 prop_binCROS_electron2018_bin12
217 prop_binCRTT_muon2017_bin3
218 prop_binCROS_electron2018_bin17
219 prop_binCROS_muon2018_bin6
220 prop_binCRTT_muon2018_bin8
221 prop_binCROS_muon2016M_bin6
222 prop_binCRTT_electron2016M_bin1
223 prop_binSR_muon2016M_bin13_WpWpJJ_QCD
224 prop_binCROS_muon2016M_bin7
225 prop_binSR_electron2017_bin14_WpWpJJ_QCD
226 pdf_Tot_WrongSign
227 prop_binCRF_electron2018_bin0
228 prop_binSR_electron2016M_bin14_TVX
229 prop_binSR_muon2018_bin0
230 prop_binSR_electron2017_bin8_TTTo2L2Nu
231 prop_binSR_electron2016M_bin14_WpWpJJ_QCD
232 prop_binCRF_electron2017_bin0
233 ISR_TTdilep
234 prop_binSR_muon2016M_bin14_WpWpJJ_QCD
235 prop_binSR_muon2016M_bin0
236 prop_binCRTT_electron2018_bin2
237 prop_binCROS_electron2016M_bin8
238 QCDScale_WrongSign
239 prop_binCRTT_electron2018_bin6
240 prop_binCROS_muon2017_bin6
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

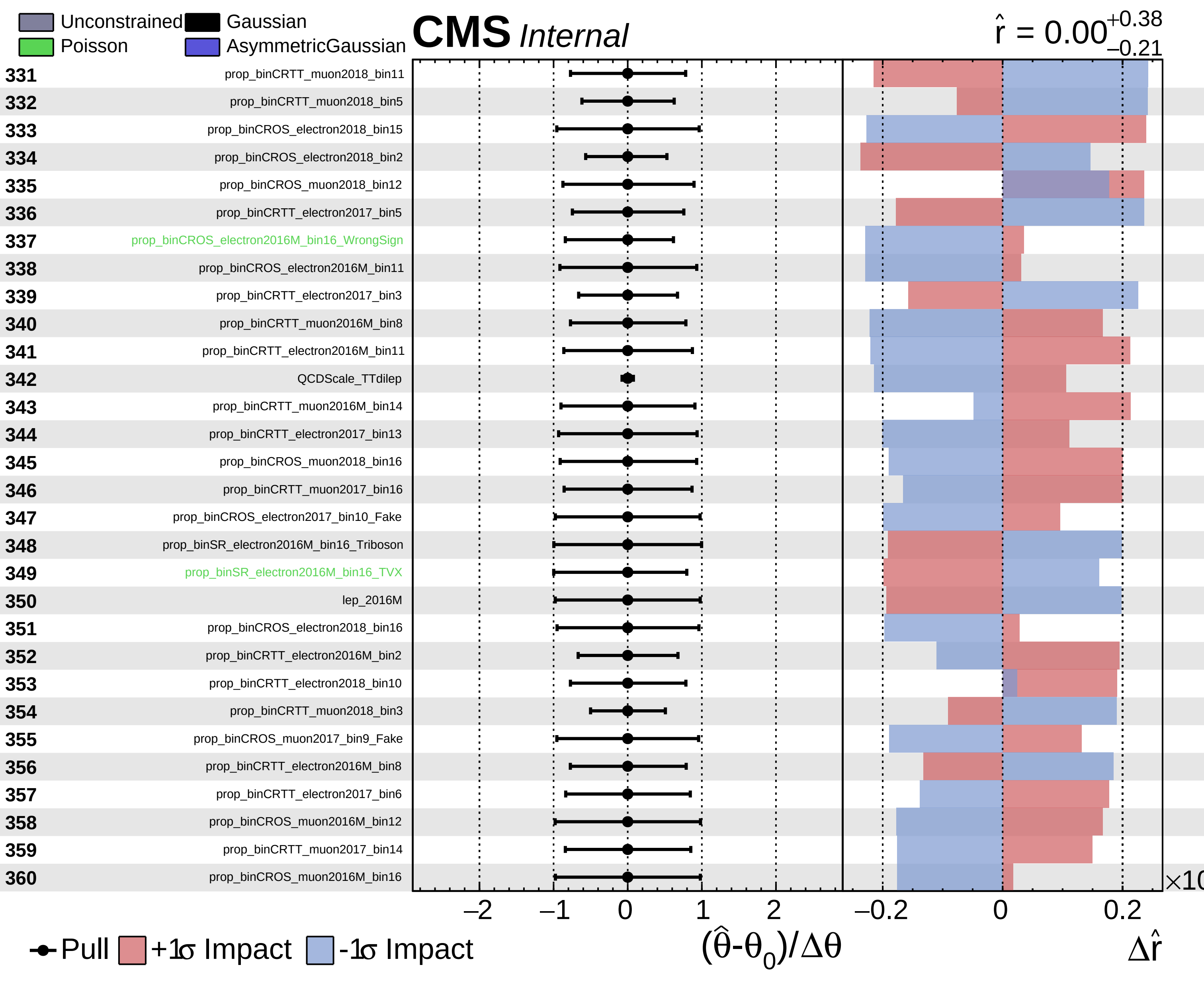

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.21}$
241 prop_binSR_muon2016M_bin13_TVX
242 prop_binCROS_electron2017_bin6
243 prop_binCROS_electron2017_bin4
244 prop_binCROS_muon2018_bin15
245 prop_binCROS_electron2017_bin2
246 prop_binCRTT_electron2016M_bin3
247 prop_binCROS_muon2017_bin2
248 prop_binCROS_muon2018_bin13
249 prop_binCRTT_electron2016M_bin7
250 prop_binSR_muon2016M_bin14_Triboson
251 prop_binCROS_electron2016M_bin5
252 prop_binCROS_electron2017_bin10_WrongSign
253 prop_binCROS_muon2018_bin9_WrongSign
254 prop_binCROS_muon2018_bin4
255 prop_binCRTT_electron2018_bin7
256 prop_binSR_muon2017_bin17_WrongSign
257 prop_binCRTT_electron2017_bin4
258 prop_binSR_muon2016M_bin13_WrongSign
259 prop_binSR_electron2016M_bin0
260 prop_binSR_electron2018_bin0
261 prop_binSR_electron2016M_bin14_WrongSign
262 prop_binCRTT_electron2018_bin9
263 prop_binCRTT_electron2018_bin5
264 prop_binSR_electron2016M_bin14_Triboson
265 prop_binCROS_electron2018_bin5
266 prop_binCRTT_muon2017_bin0
267 prop_binCROS_electron2016M_bin6
268 prop_binCRTT_muon2018_bin12
269 prop_binCROS_electron2017_bin14
270 prop_binCRTT_muon2018_bin6
−2 −1 0 1 2
−0.5 0 0.5
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



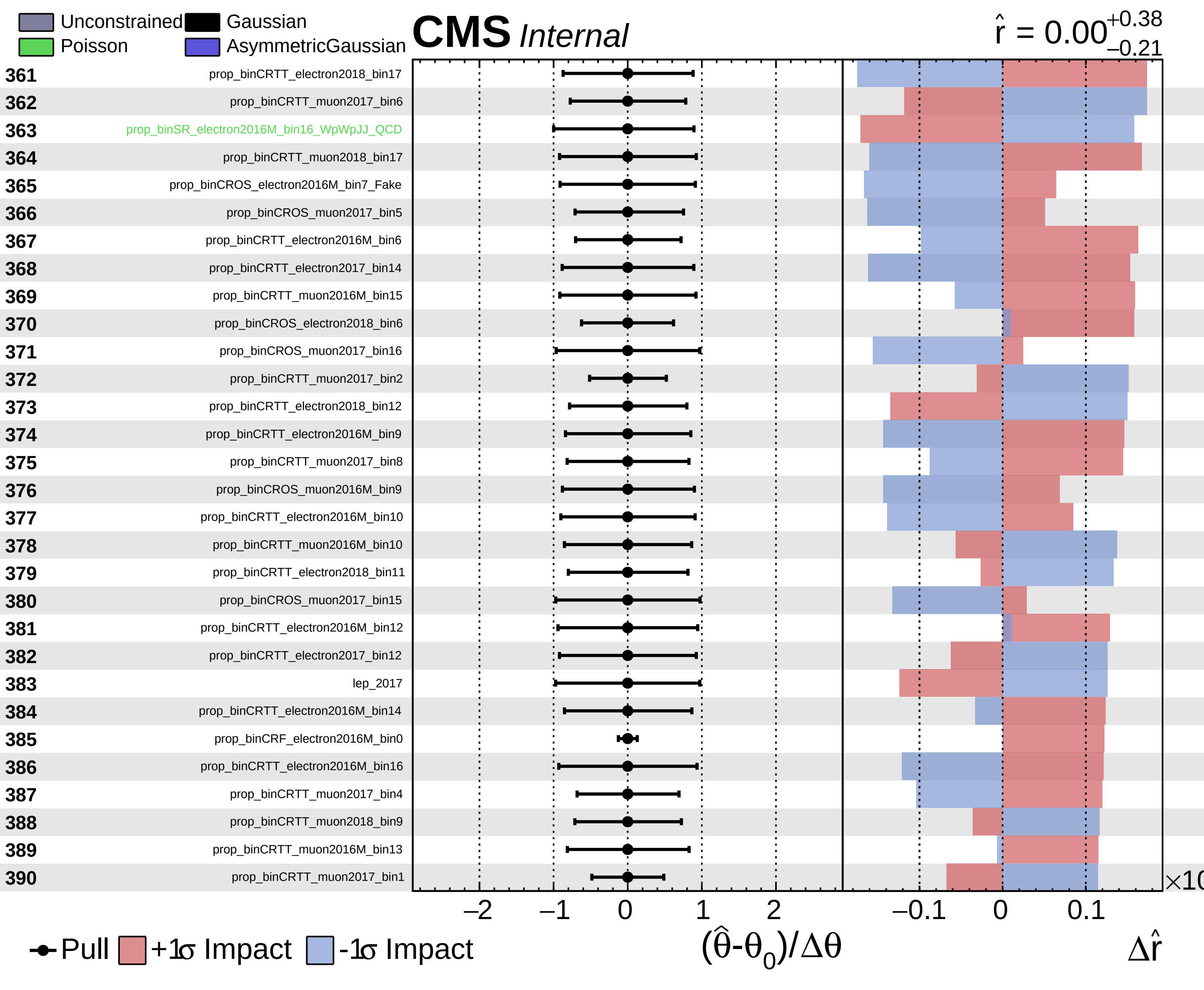
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.21}$
271 prop_binCRTT_muon2017_bin11
272 prop_binSR_muon2016M_bin13_Triboson
273 prop_binSR_electron2017_bin14_WrongSign
274 prop_binCRTT_muon2016M_bin4
275 prop_binCROS_electron2016M_bin4
276 prop_binCRTT_muon2018_bin10
277 prop_binCROS_muon2018_bin8
278 prop_binCRTT_electron2016M_bin5
279 prop_binCROS_muon2017_bin10
280 prop_binCRTT_muon2016M_bin1
281 prop_binCROS_electron2017_bin7_Fake
282 prop_binCROS_electron2017_bin12
283 prop_binSR_electron2016M_bin17_Triboson
284 prop_binSR_electron2016M_bin17_WpWpJJ_QCD
285 prop_binSR_muon2016M_bin14_TVX
286 prop_binCROS_electron2018_bin10
287 prop_binCROS_electron2018_bin4
288 prop_binCRTT_electron2017_bin1
289 prop_binSR_electron2017_bin8_WpWpJJ_QCD
290 prop_binCRTT_electron2017_bin2
291 prop_binCRTT_muon2016M_bin3
292 prop_binCRTT_muon2018_bin15
293 prop_binCRTT_muon2017_bin12
294 prop_binCRTT_muon2018_bin4
295 prop_binCROS_electron2018_bin13
296 prop_binCRTT_muon2016M_bin7
297 prop_binCRTT_muon2018_bin13
298 prop_binCRTT_electron2017_bin11
299 prop_binCROS_muon2016M_bin14
300 prop_binCROS_muon2017_bin9_WrongSign
−2
−1
0
1
2
−0.4
−0.2
0
0.2
0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



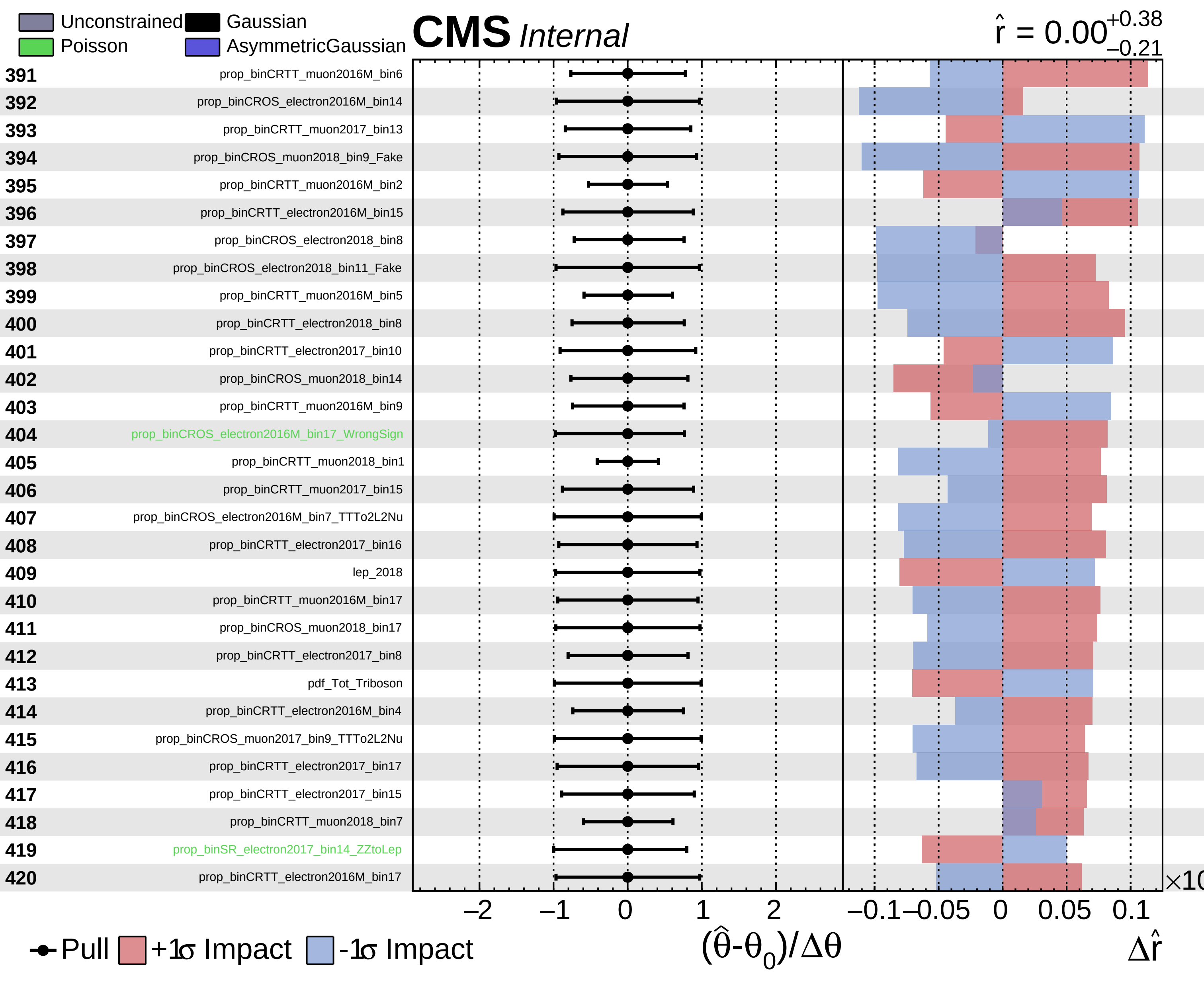
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.21}$
301 prop_binCRTT_electron2018_bin4
302 prop_binCROS_muon2016M_bin10
303 prop_binCROS_muon2017_bin11
304 prop_binCROS_electron2018_bin11_WrongSign
305 prop_binCRTT_muon2017_bin9
306 prop_binCROS_electron2017_bin13
307 prop_binSR_muon2016M_bin14_WrongSign
308 prop_binCROS_electron2016M_bin10
309 prop_binCRTT_electron2016M_bin13
310 prop_binCROS_electron2017_bin8_WrongSign
311 prop_binCRTT_electron2017_bin9
312 prop_binSR_electron2017_bin14_Triboson
313 prop_binCROS_electron2017_bin8_Fake
314 prop_binSR_electron2017_bin0
315 prop_binCRTT_muon2017_bin5
316 prop_binCRTT_muon2016M_bin12
317 prop_binCROS_electron2016M_bin12
318 prop_binCRTT_muon2018_bin16
319 prop_binCROS_muon2016M_bin11
320 prop_binCRTT_muon2017_bin10
321 prop_binCROS_muon2018_bin11
322 prop_binCROS_muon2017_bin7
323 prop_binCROS_muon2017_bin4_Fake
324 prop_binCRTT_electron2018_bin15
325 prop_binCRTT_electron2018_bin16
326 prop_binCROS_muon2017_bin4_WrongSign
327 prop_binCRTT_muon2016M_bin11
328 prop_binSR_electron2017_bin8_TVX
329 prop_binCROS_muon2017_bin12
330 FSR_TTdilep
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



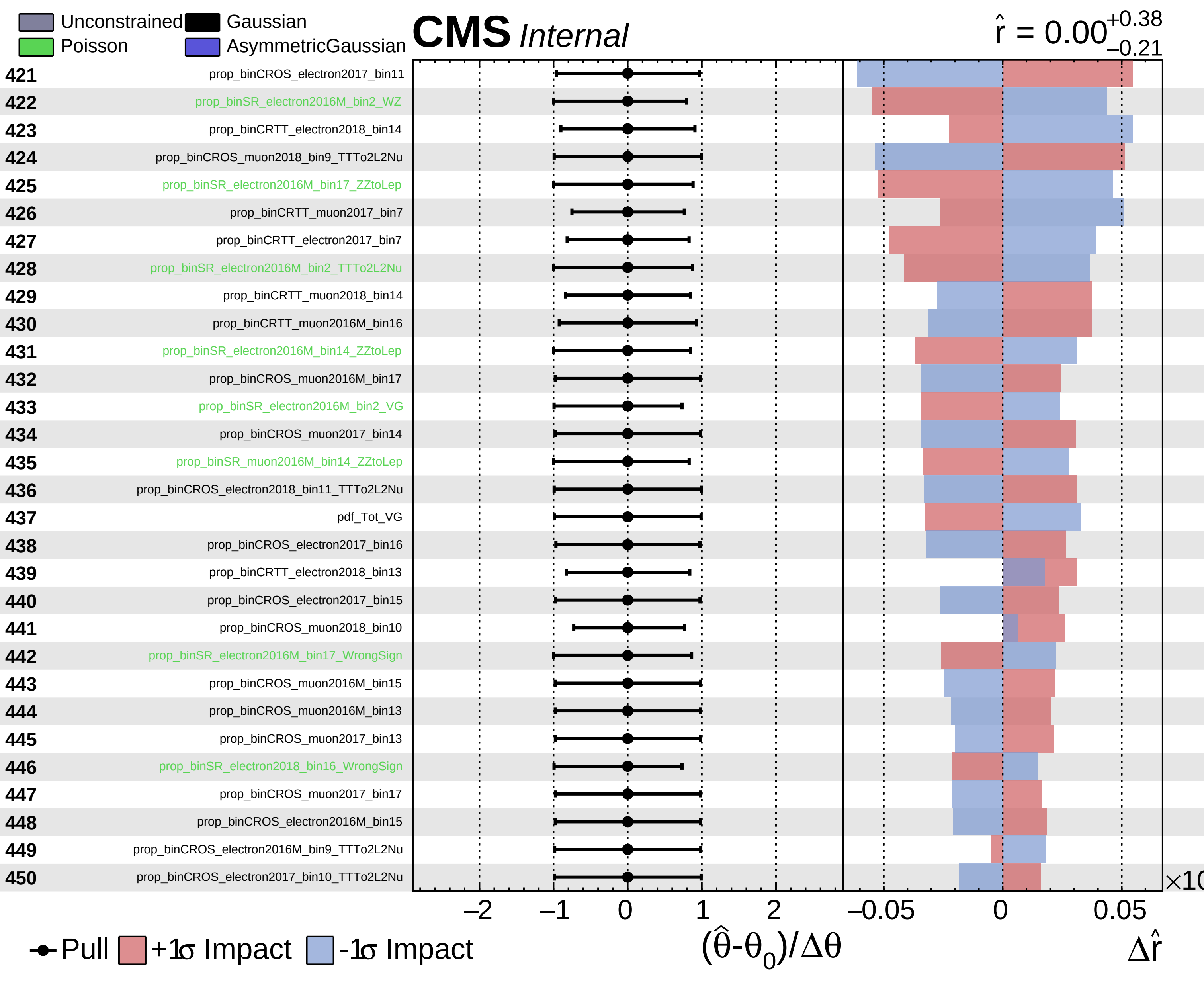
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.21}$
331 prop_binCRTT_muon2018_bin11
332 prop_binCRTT_muon2018_bin5
333 prop_binCROS_electron2018_bin15
334 prop_binCROS_electron2018_bin2
335 prop_binCROS_muon2018_bin12
336 prop_binCRTT_electron2017_bin5
337 prop_binCROS_electron2016M_bin16_WrongSign
338 prop_binCROS_electron2016M_bin11
339 prop_binCRTT_electron2017_bin3
340 prop_binCRTT_muon2016M_bin8
341 prop_binCRTT_electron2016M_bin11
342 QCDScale_TTdilep
343 prop_binCRTT_muon2016M_bin14
344 prop_binCRTT_electron2017_bin13
345 prop_binCROS_muon2018_bin16
346 prop_binCRTT_muon2017_bin16
347 prop_binCROS_electron2017_bin10_Fake
348 prop_binSR_electron2016M_bin16_Triboson
349 prop_binSR_electron2016M_bin16_TVX
350 lep_2016M
351 prop_binCROS_electron2018_bin16
352 prop_binCRTT_electron2016M_bin2
353 prop_binCRTT_electron2018_bin10
354 prop_binCRTT_muon2018_bin3
355 prop_binCROS_muon2017_bin9_Fake
356 prop_binCRTT_electron2016M_bin8
357 prop_binCRTT_electron2017_bin6
358 prop_binCROS_muon2016M_bin12
359 prop_binCRTT_muon2017_bin14
360 prop_binCROS_muon2016M_bin16
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



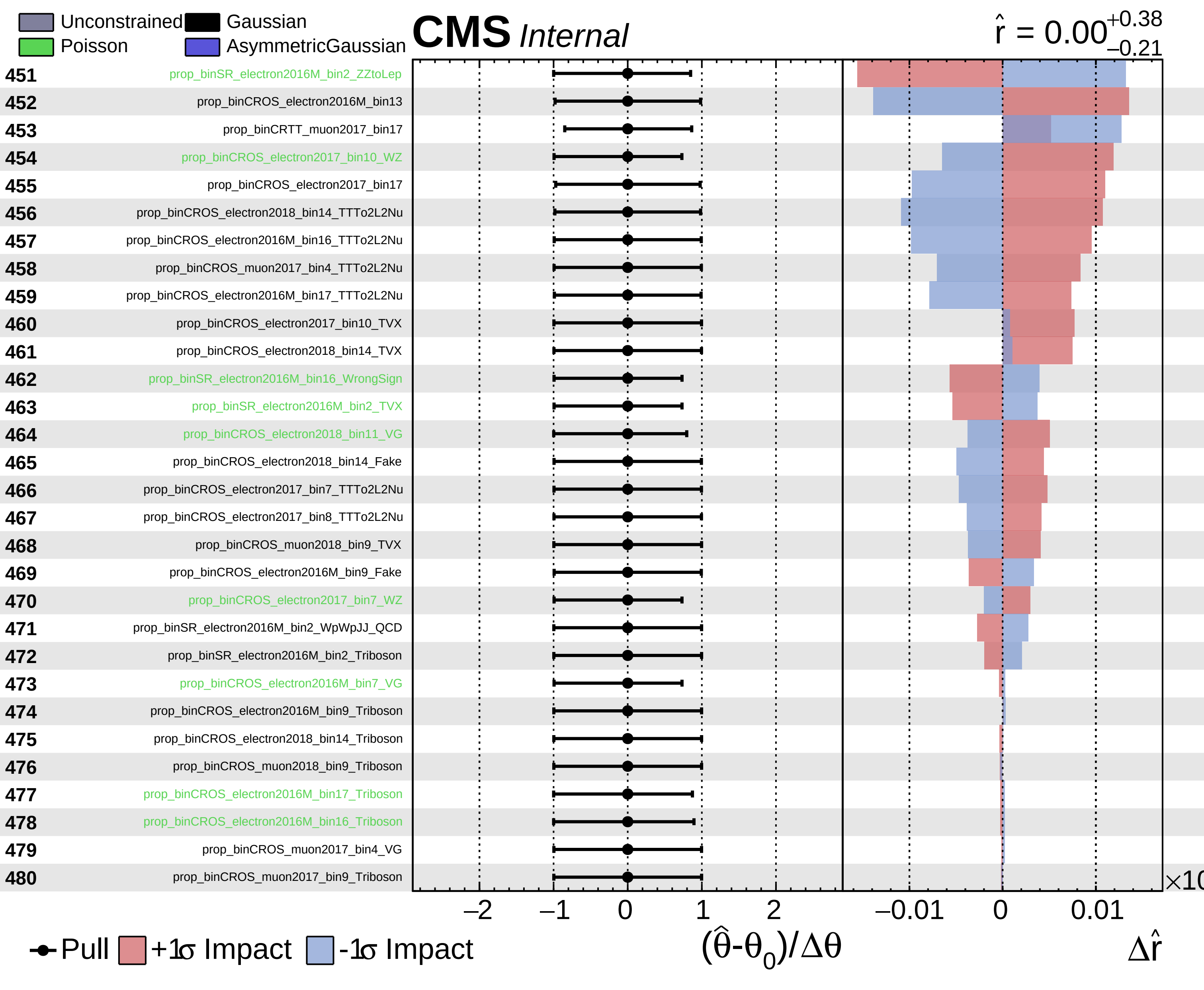
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.21}$
361 prop_binCRTT_electron2018_bin17
362 prop_binCRTT_muon2017_bin6
363 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
364 prop_binCRTT_muon2018_bin17
365 prop_binCROS_electron2016M_bin7_Fake
366 prop_binCROS_muon2017_bin5
367 prop_binCRTT_electron2016M_bin6
368 prop_binCRTT_electron2017_bin14
369 prop_binCRTT_muon2016M_bin15
370 prop_binCROS_electron2018_bin6
371 prop_binCROS_muon2017_bin16
372 prop_binCRTT_muon2017_bin2
373 prop_binCRTT_electron2018_bin12
374 prop_binCRTT_electron2016M_bin9
375 prop_binCRTT_muon2017_bin8
376 prop_binCROS_muon2016M_bin9
377 prop_binCRTT_electron2016M_bin10
378 prop_binCRTT_muon2016M_bin10
379 prop_binCRTT_electron2018_bin11
380 prop_binCROS_muon2017_bin15
381 prop_binCRTT_electron2016M_bin12
382 prop_binCRTT_electron2017_bin12
383 lep_2017
384 prop_binCRTT_electron2016M_bin14
385 prop_binCRF_electron2016M_bin0
386 prop_binCRTT_electron2016M_bin16
387 prop_binCRTT_muon2017_bin4
388 prop_binCRTT_muon2018_bin9
389 prop_binCRTT_muon2016M_bin13
390 prop_binCRTT_muon2017_bin1
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



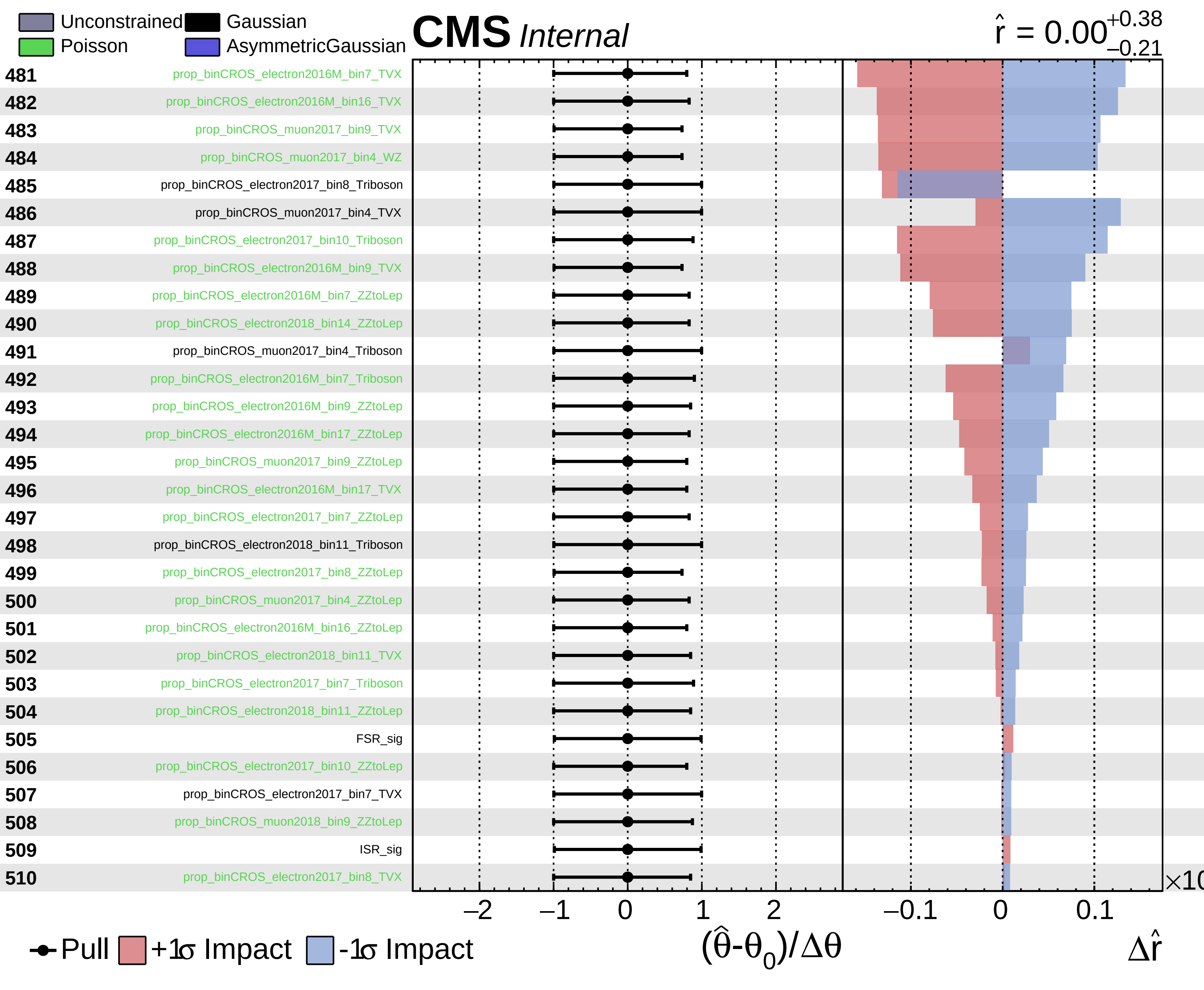
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.21}$
391 prop_binCRTT_muon2016M_bin6
392 prop_binCROS_electron2016M_bin14
393 prop_binCRTT_muon2017_bin13
394 prop_binCROS_muon2018_bin9_Fake
395 prop_binCRTT_muon2016M_bin2
396 prop_binCRTT_electron2016M_bin15
397 prop_binCROS_electron2018_bin8
398 prop_binCROS_electron2018_bin11_Fake
399 prop_binCRTT_muon2016M_bin5
400 prop_binCRTT_electron2018_bin8
401 prop_binCRTT_electron2017_bin10
402 prop_binCROS_muon2018_bin14
403 prop_binCRTT_muon2016M_bin9
404 prop_binCROS_electron2016M_bin17_WrongSign
405 prop_binCRTT_muon2018_bin1
406 prop_binCRTT_muon2017_bin15
407 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
408 prop_binCRTT_electron2017_bin16
409 lep_2018
410 prop_binCRTT_muon2016M_bin17
411 prop_binCROS_muon2018_bin17
412 prop_binCRTT_electron2017_bin8
413 pdf_Tot_Triboson
414 prop_binCRTT_electron2016M_bin4
415 prop_binCROS_muon2017_bin9_TTTo2L2Nu
416 prop_binCRTT_electron2017_bin17
417 prop_binCRTT_electron2017_bin15
418 prop_binCRTT_muon2018_bin7
419 prop_binSR_electron2017_bin14_ZZtoLep
420 prop_binCRTT_electron2016M_bin17
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



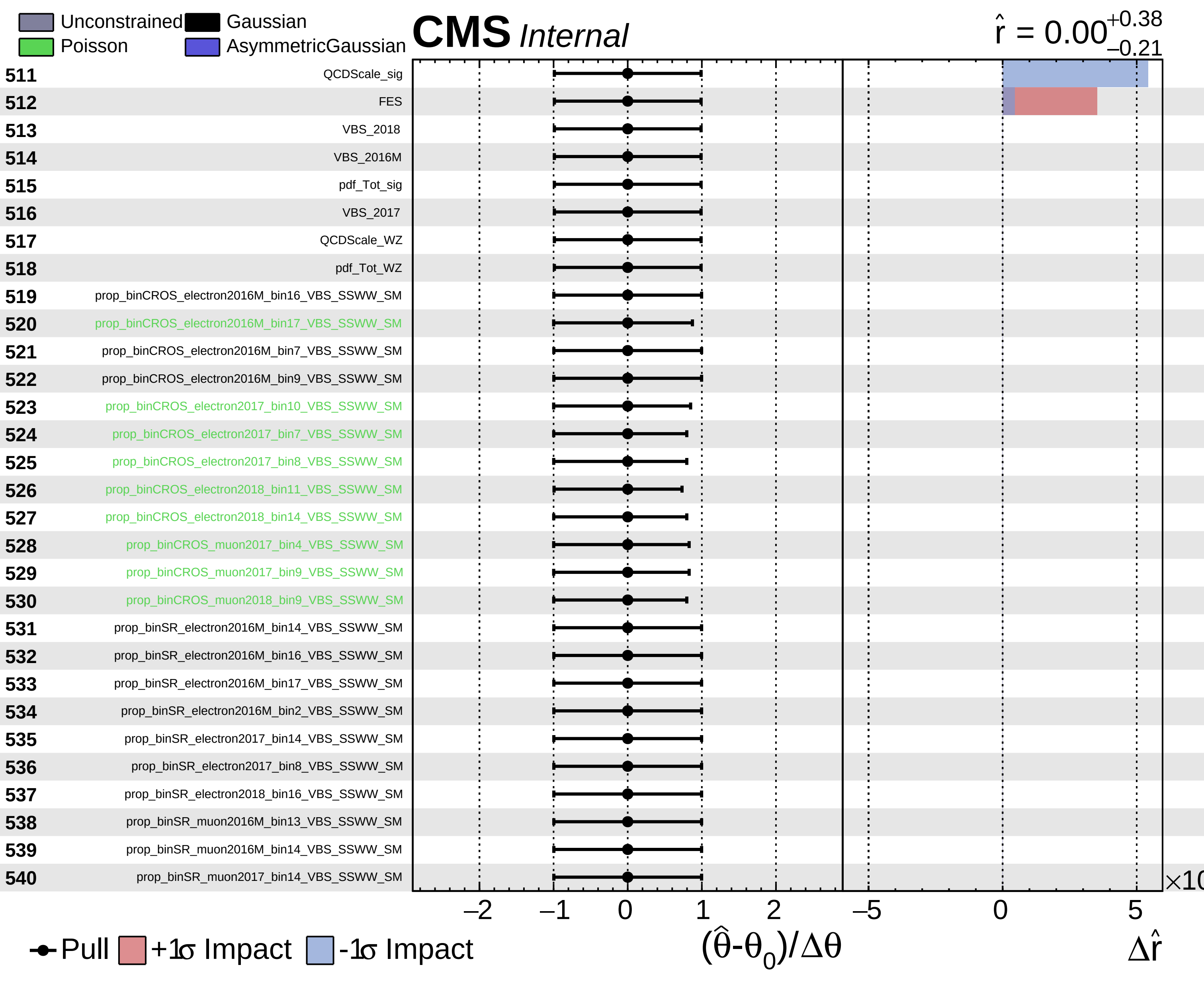
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.38 −0.21
421 prop_binCROS_electron2017_bin11
422 prop_binSR_electron2016M_bin2_WZ
423 prop_binCRTT_electron2018_bin14
424 prop_binCROS_muon2018_bin9_TTTo2L2Nu
425 prop_binSR_electron2016M_bin17_ZZtoLep
426 prop_binCRTT_muon2017_bin7
427 prop_binCRTT_electron2017_bin7
428 prop_binSR_electron2016M_bin2_TTTo2L2Nu
429 prop_binCRTT_muon2018_bin14
430 prop_binCRTT_muon2016M_bin16
431 prop_binSR_electron2016M_bin14_ZZtoLep
432 prop_binCROS_muon2016M_bin17
433 prop_binSR_electron2016M_bin2_VG
434 prop_binCROS_muon2017_bin14
435 prop_binSR_muon2016M_bin14_ZZtoLep
436 prop_binCROS_electron2018_bin11_TTTo2L2Nu
437 pdf_Tot_VG
438 prop_binCROS_electron2017_bin16
439 prop_binCRTT_electron2018_bin13
440 prop_binCROS_electron2017_bin15
441 prop_binCROS_muon2018_bin10
442 prop_binSR_electron2016M_bin17_WrongSign
443 prop_binCROS_muon2016M_bin15
444 prop_binCROS_muon2016M_bin13
445 prop_binCROS_muon2017_bin13
446 prop_binSR_electron2018_bin16_WrongSign
447 prop_binCROS_muon2017_bin17
448 prop_binCROS_electron2016M_bin15
449 prop_binCROS_electron2016M_bin9_TTTo2L2Nu
450 prop_binCROS_electron2017_bin10_TTTo2L2Nu
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

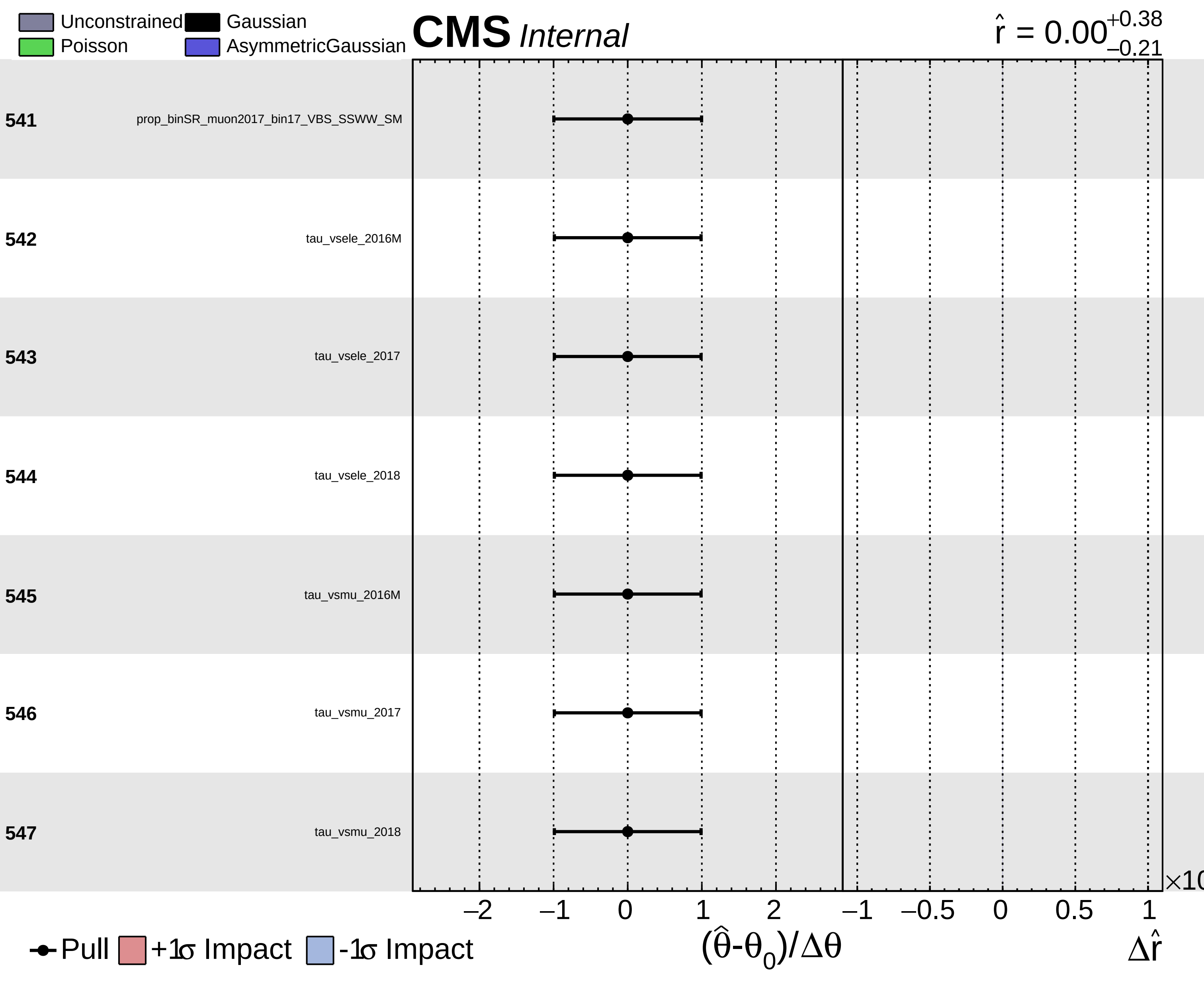


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.21}$
451 prop_binSR_electron2016M_bin2_ZZtoLep
452 prop_binCROS_electron2016M_bin13
453 prop_binCRTT_muon2017_bin17
454 prop_binCROS_electron2017_bin10_WZ
455 prop_binCROS_electron2017_bin17
456 prop_binCROS_electron2018_bin14_TTTo2L2Nu
457 prop_binCROS_electron2016M_bin16_TTTo2L2Nu
458 prop_binCROS_muon2017_bin4_TTTo2L2Nu
459 prop_binCROS_electron2016M_bin17_TTTo2L2Nu
460 prop_binCROS_electron2017_bin10_TVX
461 prop_binCROS_electron2018_bin14_TVX
462 prop_binSR_electron2016M_bin16_WrongSign
463 prop_binSR_electron2016M_bin2_TVX
464 prop_binCROS_electron2018_bin11_VG
465 prop_binCROS_electron2018_bin14_Fake
466 prop_binCROS_electron2017_bin7_TTTo2L2Nu
467 prop_binCROS_electron2017_bin8_TTTo2L2Nu
468 prop_binCROS_muon2018_bin9_TVX
469 prop_binCROS_electron2016M_bin9_Fake
470 prop_binCROS_electron2017_bin7_WZ
471 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
472 prop_binSR_electron2016M_bin2_Triboson
473 prop_binCROS_electron2016M_bin7_VG
474 prop_binCROS_electron2016M_bin9_Triboson
475 prop_binCROS_electron2018_bin14_Triboson
476 prop_binCROS_muon2018_bin9_Triboson
477 prop_binCROS_electron2016M_bin17_Triboson
478 prop_binCROS_electron2016M_bin16_Triboson
479 prop_binCROS_muon2017_bin4_VG
480 prop_binCROS_muon2017_bin9_Triboson
−2 −1 0 1 2
−0.01 0 0.01
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.21}$
481 prop_binCROS_electron2016M_bin7_TVX
482 prop_binCROS_electron2016M_bin16_TVX
483 prop_binCROS_muon2017_bin9_TVX
484 prop_binCROS_muon2017_bin4_WZ
485 prop_binCROS_electron2017_bin8_Triboson
486 prop_binCROS_muon2017_bin4_TVX
487 prop_binCROS_electron2017_bin10_Triboson
488 prop_binCROS_electron2016M_bin9_TVX
489 prop_binCROS_electron2016M_bin7_ZZtoLep
490 prop_binCROS_electron2018_bin14_ZZtoLep
491 prop_binCROS_muon2017_bin4_Triboson
492 prop_binCROS_electron2016M_bin7_Triboson
493 prop_binCROS_electron2016M_bin9_ZZtoLep
494 prop_binCROS_electron2016M_bin17_ZZtoLep
495 prop_binCROS_muon2017_bin9_ZZtoLep
496 prop_binCROS_electron2016M_bin17_TVX
497 prop_binCROS_electron2017_bin7_ZZtoLep
498 prop_binCROS_electron2018_bin11_Triboson
499 prop_binCROS_electron2017_bin8_ZZtoLep
500 prop_binCROS_muon2017_bin4_ZZtoLep
501 prop_binCROS_electron2016M_bin16_ZZtoLep
502 prop_binCROS_electron2018_bin11_TVX
503 prop_binCROS_electron2017_bin7_Triboson
504 prop_binCROS_electron2018_bin11_ZZtoLep
505 FSR_sig
506 prop_binCROS_electron2017_bin10_ZZtoLep
507 prop_binCROS_electron2017_bin7_TVX
508 prop_binCROS_muon2018_bin9_ZZtoLep
509 ISR_sig
510 prop_binCROS_electron2017_bin8_TVX
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.21}$
511 QCDScale_sig
512 FES
513 VBS_2018
514 VBS_2016M
515 pdf_Tot_sig
516 VBS_2017
517 QCDScale_WZ
518 pdf_Tot_WZ
519 prop_binCROS_electron2016M_bin16_VBS_SSWW_SM
520 prop_binCROS_electron2016M_bin17_VBS_SSWW_SM
521 prop_binCROS_electron2016M_bin7_VBS_SSWW_SM
522 prop_binCROS_electron2016M_bin9_VBS_SSWW_SM
523 prop_binCROS_electron2017_bin10_VBS_SSWW_SM
524 prop_binCROS_electron2017_bin7_VBS_SSWW_SM
525 prop_binCROS_electron2017_bin8_VBS_SSWW_SM
526 prop_binCROS_electron2018_bin11_VBS_SSWW_SM
527 prop_binCROS_electron2018_bin14_VBS_SSWW_SM
528 prop_binCROS_muon2017_bin4_VBS_SSWW_SM
529 prop_binCROS_muon2017_bin9_VBS_SSWW_SM
530 prop_binCROS_muon2018_bin9_VBS_SSWW_SM
531 prop_binSR_electron2016M_bin14_VBS_SSWW_SM
532 prop_binSR_electron2016M_bin16_VBS_SSWW_SM
533 prop_binSR_electron2016M_bin17_VBS_SSWW_SM
534 prop_binSR_electron2016M_bin2_VBS_SSWW_SM
535 prop_binSR_electron2017_bin14_VBS_SSWW_SM
536 prop_binSR_electron2017_bin8_VBS_SSWW_SM
537 prop_binSR_electron2018_bin16_VBS_SSWW_SM
538 prop_binSR_muon2016M_bin13_VBS_SSWW_SM
539 prop_binSR_muon2016M_bin14_VBS_SSWW_SM
540 prop_binSR_muon2017_bin14_VBS_SSWW_SM
−2 −1 0 1 2
−5 0 5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.38 −0.21
541 prop_binSR_muon2017_bin17_VBS_SSWW_SM
542 tau_vsele_2016M
543 tau_vsele_2017
544 tau_vsele_2018
545 tau_vsmu_2016M
546 tau_vsmu_2017
547 tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂